



Sumit Tapas Chongder

Master of Technology
Quantum Technologies
Indian Institute of Technology, Jodhpur

+91-7738915928
sumitchongder960@gmail.com
linkedin.com/in/sumit-chongder
github.com/Sumitchongder

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech QT	Indian Institute of Technology, Jodhpur	9.25 (Rank Holder)	2025-2027
B.Tech, CSE	MIT Art, Design and Technology University	8.92 (Rank Holder)	2024
Senior Secondary	Pace Junior Science College, Maharashtra	81.69%	2020
Secondary	St. Francis High School, Maharashtra	94% (Rank Holder)	2018

EXPERIENCE

- QNu Labs** Remote
Summer Research Intern May 2026 – Jul 2026
 - Engineered and benchmarked industry-grade **CSPRNGs**, hardware **TRNGs** (microprocessors & camera sensors), and homodyne-detection **QRNGs**; validated entropy quality metrics across a **4.15 GB** multi-source binary dataset using *NIST SP 800-22*, *NIST SP 800-90B*, *Dieharder*, and *ENT* test suites.
 - Designed and deployed a hierarchical randomness source attribution pipeline extracting **64 statistical features** per window, **via** calibrated ensembles (**CatBoost**, **XGBoost**, **LightGBM**) yielding **81.21%** mean confidence.
- Africa Quantum Consortium** Remote
Mentor – Quantum Error Correction & Mitigation (QEM) Track Dec 2025 – Jan 2026
 - Mentored **5 interns** (Pan-African Quantum Challenge) on **automated QEC code construction**, **CUDA-Q** simulation, and ML decoders for **topological**, **stabilizer**, & **QLDPC** codes.
 - Co-developed a two-tier framework routing 3.3%–6.2% of surface code ($d \in \{3, 5, 7, 11\}$) syndromes to **PyMatching MWPM**, boosting accuracy to **99.81%** under **fault-tolerant circuit-level noise**.
- Fortinet** Remote
Network Security Associate (Internship) Sep 2023 – Nov 2023
 - Leveraged **FCA & FCF Cybersecurity** knowledge to analyze network traffic and manage security policies on **Fortinet devices**, mitigating threats and ensuring 100% compliance with multi-layer **security protocols**.
 - Investigated security breaches and orchestrated **incident resolution** workflows using advanced detection & response methodologies, reducing threat mitigation time by 35% and securing critical data infrastructure.
- Amazon Web Services (AWS)** Remote
Cloud Associate (Internship) May 2023 – Jul 2023
 - Architected **secure AWS infrastructure** (EC2, S3, VPC) using Cloud Foundations & Architecting principles, achieving 99.99% **high-availability** and ensuring strict multi-layer network isolation.
 - Built **automated CI/CD pipelines** via CodePipeline, CloudFormation, and Lambda, reducing deployment time by 40% and delivering highly **scalable, cost-optimized solutions**.
- Microsoft (Future Ready Talent Program)** Remote
Software Engineer (Internship) Feb 2023 – Apr 2023
 - Developed responsive apps deployed via **Azure Static Web Apps** and **App Services**, integrating **Bot & Cognitive Services** to achieve 99.9% **high availability** and intelligent text analytics.
 - Built secure **Azure Storage** architectures and automated **CI/CD pipelines**, leveraging **Azure Monitor** and Application Insights to reduce system downtime by 25%.

PUBLICATIONS & PREPRINTS

- QR-SPPS: Quantum-Native Retail Supply Chain Risk Simulation via VQE, ADAPT-VQE Counterfactual Policy Ranking, and DOS-QPE Boltzmann Tail Risk Quantification:** Chongder, S. T., *arXiv:2604.00035* [quant-ph] DOI: [10.48550/arXiv.2604.00035](https://doi.org/10.48550/arXiv.2604.00035)
- Latency-Constrained Hardware-Aware Quantum Error Correction Co-Design with Adaptive Confidence-Gated Neural Decoding for the Rotated Surface Code:** Chongder, S. T., *arXiv:2607.05814* [quant-ph] DOI: [10.48550/arXiv.2607.05814](https://doi.org/10.48550/arXiv.2607.05814)
- Hardware-aware Low-latency Quantum Compilation with Data-driven Lightweight Error Detection for Early Fault-Tolerant Systems:** Chongder, S. T., *arXiv:2606.07666* [quant-ph] DOI: [10.48550/arXiv.2606.07666](https://doi.org/10.48550/arXiv.2606.07666)
- Chongder, S. T. **Evaluation Framework for Centralized and Decentralized Aggregation Algorithm in Federated Systems.** *Cureus Journal of Computer Science*, 2026. DOI: [10.7759/s44389-026-00028-0](https://doi.org/10.7759/s44389-026-00028-0)

INTELLECTUAL PROPERTY RIGHTS – GOVERNMENT OF INDIA (COPYRIGHTS/PATENTS)

- **Quantum Resistant Module Lattice Cryptography System** | [ROC No. SW-2026022888](#) | *27/05/2026*
- **Analytical BB84 Quantum Key Distribution (QKD) Engine** | [ROC No. SW-2026022477](#) | *03/03/2026*
- **Quantum Hyperspectral Imaging (Q-HSI) System** | [ROC No. SW-2026022430](#) | *27/02/2026*
- **System And Method For Quantum Phase Space Simulation Of Quantum States** | [ROC No. SW-2026023370](#) | *22/07/2026*
- **Quantum Native Retail Shock Propagation And Policy Stress Simulator** | *Diary No. SW-24574/2026-CO* | *Filed on 25/05/2026*

PROJECTS

- **QR-SPPS: Quantum Supply Chain Risk Simulator** | *Fujitsu Quantum Challenge 2026* | [GitHub](#)
 - Engineered a 40-qubit Ising Hamiltonian on A64FX supercomputers using QARP/MPI for zero-error VQE convergence, employing ADAPT-VQE and DOS-QPE to optimize real-time supply chain interventions.
- **Hybrid Quantum-Classical QAOA Framework** | *IIT Jodhpur* | [GitHub](#)
 - Engineered a **GPU-accelerated Qiskit Aer** framework benchmarking **12 classical SciPy optimizers** for MAX-CUT QAOA, executing deep parameter sweeps across HPC nodes to systematically mitigate local minima.
- **Analytical BB84 QKD Engine** | *Qiskit Fall Fest Winner* | [GitHub](#)
 - Developed a dual-backend Qiskit & Streamlit QKD simulator featuring vectorized probability models, multi-strategy eavesdropping detection dashboards, and automated privacy amplification under a strict 11% QBER abort threshold.
- **Quantum Phase Space Simulator** | *IIT Jodhpur* | [GitHub](#)
 - Interactive Streamlit dashboard for continuous-variable quantum information: Wigner functions, density matrices, Husimi Q, GBS, and CV-QML for 8 quantum states using QuTiP, Strawberry Fields, and PennyLane.
- **Quantum-Resistant Module Lattice Cryptography** | [GitHub](#)
 - Engineered crypto-agile Rust wallets integrating Kyber-1024 KEM, Dilithium5 signatures, and XChaCha20-Poly1305 symmetric envelopes to secure data storage with tamper-proof transaction verification.
- **Quantum Hyperspectral Imaging (Q-HSI)** | [GitHub](#)
 - Developed a hybrid CNN-VQC ensemble using PennyLane to achieve 83.5% diagnostic accuracy, accelerating quantum training throughput 13× faster than classical baselines.

SKILLS

Programming Languages	Python, CUDA-Q, Rust, C++, SQL
Quantum Frameworks	Qiskit, PennyLane, Strawberry Fields, QuTiP, Cirq
ML Libraries	TensorFlow, PyTorch, Scikit-learn, Keras, NumPy, Pandas, Matplotlib
Cloud & DevOps	AWS (EC2, S3, Lambda, CodePipeline), Microsoft Azure (App Services, Static Web Apps, Storage)
Other Tools	Git, GitHub, Streamlit, VS Code, MySQL, Jupyter Notebook

LICENSES & CERTIFICATIONS

- **Foundations of Quantum Error Correction** – IBM | *Jun 2026* | [Credential Link](#)
- **Integrating Quantum and High-Performance Computing** – IBM | *Jun 2026* | [Credential Link](#)
- **Quantum Machine Learning** – IBM | *Jun 2026* | [Credential Link](#)
- **Practical Introduction to Quantum-Safe Cryptography** – IBM | *Jun 2026* | [Credential Link](#)
- **QSilver – Diploma in Quantum Computing & Programming** – QWorld | [Credential Link](#)
- **QBronze – Diploma in Quantum Computing & Programming** – QWorld | [Credential Link](#)

HONORS, AWARDS & HACKATHONS

- **1st Place – IBM Qiskit Fall Fest IIT Jodhpur Hackathon 2025** | IBM Quantum & IIT Jodhpur | *Nov 2025*
Project: “AnalytiQ: Analytical BB84 QKD Engine” – Implemented and simulated QKD protocols with focus on security, channel integrity, and quantum communication workflows using Qiskit. | [Certificate Link](#)
- **Top 50 – MIT iQuHack 2026 (Rank 46 Worldwide)** | BlueQubit X MIT | *Jan 2026* | [Certificate Link](#)
- **Top 40 – Yale Peaked Hackathon 2026 (Rank 38 Worldwide)** | BlueQubit X Yale | *Apr 2026*
- **Rank Holder in Academics** – MIT Art Design and Technology University, Pune | *Sep 2024*

LEADERSHIP & VOLUNTEERING

- **Qiskit Advocate – IBM Quantum**: Selected for global program; contributing to quantum computing community.
- Student member, **Quantum Information and Computation (QIC)** group, IIT Jodhpur; engaged in quantum research discussions and seminars.
- **Campus Ambassador** – Talent Battle (May 2023 – Nov 2023): Led workshops and events; increased student participation through campus outreach.
- **Student Volunteer** – HelpAge India & Indian Red Cross Society: Contributed to community service initiatives.